

Processando config 13, seed 3, teste: Rabbit
 xx
 Starting Rabbit: nb = 16777216
 Version: TestU01 1.2.3
 xx

 HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
 Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config13_seed3.bin

smultin_MultinomialBitsOver test:

 N = 2, n = 8388608, r = 0, s = 32, L = 38,
 Sparse = TRUE

 Number of bits = n = 8388608
 Number of cells = 2^L = 274877906944
 Expected number per cell = $1 / 32768$
 EColl = $n^2 / (2k)$ = 128
 Hashing = TRUE

 Collision test
 CollisionOver: density = $n / k = 1 / 32768.00$
 Expected number of collisions = $\mu = 128.00$

 Results of CollisionOver test:

POISSON approximation	:	
Expected number of collisions = $N \cdot \mu$	:	256.00
Observed number of collisions	:	241
p-value of test	:	0.82

 Total number of cells containing j balls

j = 0	:	549739036913
j = 1	:	16776734
j = 2	:	241
j = 3	:	0
j = 4	:	0
j = 5	:	0

 CPU time used : 00:00:04.45

Generator state:
 16777216 bits have been read.

 HOST = matheus-IdeaPad-3-15ALC6, Linux

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snpair_ClosePairsBitMatch test:

N = 1, n = 262144, r = 0, t = 2

Max of all bit distances : 17.00
p-value of test : 0.39

CPU time used : 00:00:00.05

Generator state:
33554432 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
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snpair_ClosePairsBitMatch test:

N = 1, n = 131072, r = 0, t = 4

Max of all bit distances : 8.00
p-value of test : 0.12

CPU time used : 00:00:00.07

Generator state:
50331648 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config13_seed3.bin

svaria_AppearanceSpacings test:

N = 1, Q = 524288, K = 524288, r = 0, s = 30, L = 15

Sequences of $n = (K + Q)L = 15728640$ bits
Q = 524288 initialization blocks
K = 524288 blocks for the test
the blocks have L = 15 bits

Normal statistic : -1.42
p-value of test : 0.92

CPU time used : 00:00:00.00

Generator state:
67108864 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
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scomp_LinearComp test:

N = 1, n = 18176, r = 0, s = 32

Number of degrees of freedom : 7
Chi2 statistic for size of jumps : 4.70
p-value of test : 0.70

Normal statistic for number of jumps : -1.41
p-value of test : 0.92

CPU time used : 00:00:00.18

Generator state:
67127040 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config13_seed3.bin

scomp_LempelZiv test:

N = 1, n = 16777216, r = 0, s = 32, k = 24

Normal statistic : 1.23
p-value of test : 0.11

CPU time used : 00:00:00.42

Generator state:

83904256 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config13_seed3.bin

sspectral_Fourier1 test:

N = 1, n = 1048576, r = 0, s = 32, k = 20

Normal statistic : 1.66
p-value of test : 0.05

CPU time used : 00:00:00.19

Generator state:

84952832 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config13_seed3.bin

sspectral_Fourier3 test:

N = 8065, n = 2048, r = 0, s = 32, k = 11

Kolmogorov-Smirnov+ statistic = D+ : 0.050
p-value of test : 0.08

Kolmogorov-Smirnov- statistic = D- : 8.43e-3
p-value of test : 0.92

Anderson-Darling statistic = A2 : 1.23
p-value of test : 0.25

CPU time used : 00:00:00.12

Generator state:

101728032 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config13_seed3.bin

sstring_LongestHeadRun test:

N = 1, n = 60, r = 0, s = 32, L = 279616

Number of degrees of freedom : 3
Chi-square statistic : 0.49
p-value of test : 0.92

Global longest run of 1 : 24.00
p-value of test : 0.39

CPU time used : 00:00:00.06

Generator state:
118504992 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config13_seed3.bin

sstring_PeriodsInStrings test:

N = 1, n = 524288, r = 0, s = 31

Number of degrees of freedom : 94
Chi-square statistic : 99.24
p-value of test : 0.34

CPU time used : 00:00:00.02

Generator state:
135282208 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config13_seed3.bin

sstring_HammingWeight test:

N = 1, n = 524288, r = 0, s = 32, L = 32

Number of degrees of freedom : 22
Chi-square statistic : 11.63
p-value of test : 0.96

CPU time used : 00:00:00.00

Generator state:
152059424 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config13_seed3.bin

sstring_HammingCorr test:

N = 1, n = 524288, r = 0, s = 32, L = 32

Normal statistic : 0.78
p-value of test : 0.22

CPU time used : 00:00:00.00

Generator state:
168836640 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config13_seed3.bin

sstring_HammingCorr test:

N = 1, n = 262144, r = 0, s = 32, L = 64

Normal statistic : 0.87
p-value of test : 0.19

CPU time used : 00:00:00.00

Generator state:
185613856 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config13_seed3.bin

sstring_HammingCorr test:

N = 1, n = 131072, r = 0, s = 32, L = 128

Normal statistic : 0.66
p-value of test : 0.25

CPU time used : 00:00:00.00

Generator state:
202391072 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config13_seed3.bin

sstring_HammingIndep test:

N = 1, n = 524288, r = 0, s = 32, L = 16, d = 0

Counters with expected numbers >= 10

Number of degrees of freedom : 177
Chi-square statistic : 183.64
p-value of test : 0.35

CPU time used : 00:00:00.00

Generator state:
219168288 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de

Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config13_seed3.bin

sstring_HammingIndep test:

N = 1, n = 262144, r = 0, s = 32, L = 32, d = 0

Counters with expected numbers >= 10

Number of degrees of freedom : 305
Chi-square statistic : 328.31
p-value of test : 0.17

CPU time used : 00:00:00.00

Generator state:

235945504 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de

Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config13_seed3.bin

sstring_HammingIndep test:

N = 1, n = 131072, r = 0, s = 32, L = 64, d = 0

Counters with expected numbers >= 10

Number of degrees of freedom : 489
Chi-square statistic : 500.09
p-value of test : 0.35

CPU time used : 00:00:00.00

Generator state:

252722720 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de

Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config13_seed3.bin

sstring_AutoCor test:

N = 1, n = 16777153, r = 0, s = 32, d = 1

```
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Normal statistic          :    0.20
p-value of test          :    0.42
-----
```

```
CPU time used            : 00:00:00.01
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Generator state:
269499904 bits have been read.
```

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*****
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HOST = matheus-IdeaPad-3-15ALC6, Linux
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ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config13_seed3.bin
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sstring_AutoCor test:
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-----
N = 1, n = 16777154, r = 0, s = 32, d = 2
-----
```

```
-----
Normal statistic          :   -0.62
p-value of test          :    0.73
-----
```

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CPU time used            : 00:00:00.01
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Generator state:
286277088 bits have been read.
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*****
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HOST = matheus-IdeaPad-3-15ALC6, Linux
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ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config13_seed3.bin
```

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sstring_Run test:
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-----
N = 1, n = 3355443, r = 0, s = 32
-----
```

```
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Total number of 1 runs: 3355447
```

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Number of degrees of freedom : 36
Chi2 statistic for number of runs : 21.75
p-value of test : 0.97
-----
```

```
-----
Total number of bits: 13426368
```

```
Normal statistic for number of bits : 0.89
p-value of test : 0.19
```

CPU time used : 00:00:00.05

Generator state:
299703456 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config13_seed3.bin

smarsa_MatrixRank test:

N = 1, n = 16384, r = 0, s = 32, L = 32, k = 32

Number of degrees of freedom : 3
Chi-square statistic : 2.48
p-value of test : 0.48

CPU time used : 00:00:00.06

Generator state:
316480672 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config13_seed3.bin

smarsa_MatrixRank test:

N = 1, n = 163, r = 0, s = 32, L = 320, k = 320

Number of degrees of freedom : 2
Chi-square statistic : 1056.73
p-value of test : eps *****

CPU time used : 00:00:00.07

Generator state:
333171872 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config13_seed3.bin

swalk_RandomWalk1 test:

N = 1, n = 131072, r = 0, s = 32, L0 = 128, L1 = 128

Test on the values of the Statistic H

Number of degrees of freedom	:	42
ChiSquare statistic	:	49.20
p-value of test	:	0.21

Test on the values of the Statistic M

Number of degrees of freedom	:	42
ChiSquare statistic	:	40.86
p-value of test	:	0.52

Test on the values of the Statistic J

Number of degrees of freedom	:	64
ChiSquare statistic	:	68.61
p-value of test	:	0.32

Test on the values of the Statistic R

Number of degrees of freedom	:	36
ChiSquare statistic	:	22.62
p-value of test	:	0.96

Test on the values of the Statistic C

Number of degrees of freedom	:	22
ChiSquare statistic	:	21.76
p-value of test	:	0.47

CPU time used : 00:00:00.05

Generator state:
349949088 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config13_seed3.bin

swalk_RandomWalk1 test:

N = 1, n = 16384, r = 0, s = 32, L0 = 1024, L1 = 1024

Test on the values of the Statistic H

Number of degrees of freedom	:	94
ChiSquare statistic	:	87.57
p-value of test	:	0.67

Test on the values of the Statistic M

Number of degrees of freedom	:	94
ChiSquare statistic	:	78.45
p-value of test	:	0.88

Test on the values of the Statistic J

Number of degrees of freedom	:	512
ChiSquare statistic	:	461.21
p-value of test	:	0.95

Test on the values of the Statistic R

Number of degrees of freedom	:	90
ChiSquare statistic	:	106.72
p-value of test	:	0.11

Test on the values of the Statistic C

Number of degrees of freedom	:	50
ChiSquare statistic	:	42.13
p-value of test	:	0.78

CPU time used : 00:00:00.05

Generator state:

366726304 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config13_seed3.bin

swalk_RandomWalk1 test:

N = 1, n = 1675, r = 0, s = 32, L0 = 10016, L1 = 10016

Test on the values of the Statistic H

Number of degrees of freedom	:	130
ChiSquare statistic	:	129.64
p-value of test	:	0.49

Test on the values of the Statistic M

Number of degrees of freedom	:	131
ChiSquare statistic	:	168.91
p-value of test	:	0.01

Test on the values of the Statistic J

Number of degrees of freedom	:	161
ChiSquare statistic	:	171.61
p-value of test	:	0.27

Test on the values of the Statistic R

Number of degrees of freedom	:	130
ChiSquare statistic	:	163.99
p-value of test	:	0.02

Test on the values of the Statistic C

Number of degrees of freedom	:	89
ChiSquare statistic	:	81.63
p-value of test	:	0.70

CPU time used : 00:00:00.04

Generator state:

383503104 bits have been read.

===== Summary results of Rabbit =====

Version:	TestU01 1.2.3
Generator:	ufile_CreateReadBin
Number of bits:	16777216
Number of statistics:	39

Total CPU time: 00:00:06.11
The following tests gave p-values outside [0.001, 0.9990]:
(eps means a value < 1.0e-300):
(eps1 means a value < 1.0e-15):

Test	p-value
-----	-----
22 MatrixRank, 320 x 320	eps
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All other tests were passed